

회귀분석

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Data science summer institute

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Multiple Linear Regression

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Agenda

1. Descriptions of data and model
2. Parameter estimation
3. Interpretation of regression of coefficients 회귀계수
4. Properties of the least squares estimators
5. Multiple correlation coefficient 중상관계수
6. Example : Supervisor Performance Data
7. Tests of hypotheses in a linear model
8. Testing all regression coefficients equal to zero
9. Testing a subset of regression coefficients equal to zero
10. Testing the equality of regression coefficients

Multiple Linear Model

The model (1) is called a multiple linear regression model.

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon \quad (1)$$

Terminology:

- ▶ Y : response
- ▶ $X_1, \dots, X_p$: predictors
- ▶ $\beta_0, \beta_1, \dots, \beta_p$: regression coefficients
- ▶ ε : random disturbance or error
 - ▶ We assume ε has a normal distribution $N(0, \sigma^2)$ for inference.

‘multiple linear regression model’은 ‘multiple regression model’으로 약해서 ‘중회귀모형’으로 번역된다.

ε 에 대한 정규성(normality) 가정은 모수에 대한 추론(inference)에 필요하다. 다른 문제, 예컨대 회귀추정량을 구하는 문제 등에서는 반드시 요구되는 가정은 아니다.

Data from a Multiple Linear Model

► Model and data

Data refer to

$$(x_{i1}, \dots, x_{ip}, y_i) \quad \text{for } i = 1, \dots, n, \quad (2)$$

where the i -th observation satisfies the linear model (1)

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \varepsilon_i.$$

► Terminology

- y_i : the i -th value of the response Y
- $x_{i1}, \dots, x_{ip}$: values of the predictors for the i -th unit
- ε_i : the error in observing y_i

Data Structure

Data (2) can be stored as follows.

Observation Number	Response Y	Predictors			
		X_1	X_2	$\cdots$	X_p
1	y_1	x_{11}	x_{12}	$\cdots$	x_{1p}
2	y_2	x_{21}	x_{22}	$\cdots$	x_{2p}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	y_n	x_{n1}	x_{n2}	$\cdots$	x_{np}

Table: Notation for the Data Used in Multiple Regression Analysis (Table 3.1)

- ▶ Meaning indices in x_{ij}
 - ▶ i = observation number
 - ▶ j = predictor number

Example: Supervisor Performance (SP) Data

- ▶ Purpose of the study

SP data show employee satisfaction with their supervisor.

- ▶ Description of variables

- ▶ Y = Overall rating of job being done by supervisor
- ▶ X_1 = Handles employee complaints
- ▶ X_2 = Does not allow special privileges
- ▶ X_3 = Opportunity to learn new things
- ▶ X_4 = Raises based on performance
- ▶ X_5 = Too critical of poor performance
- ▶ X_6 = Rate of advancing to better jobs

Generation of SP Data

Response on any item ranged from 1 (very satisfactory) through 5 (very unsatisfactory).

A dichotomous index:

- ▶ favorable response (1, 2)
- ▶ unfavorable response: (3, 4, 5)

30 departments were selected at random, where each department had approximately 35 employees and one supervisor.

The data consist of the proportions of favorable responses for each item for each department. Table 3.3 contains the collected data.

Parameter Estimation by LSE

Estimate $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_p)$ by $\hat{\boldsymbol{\beta}} = (\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p)$ which minimize the sum of squares

$$S(\mathbf{b}) = \sum_{i=1}^n (y_i - b_0 - b_1 x_{i1} - \dots - b_p x_{ip})^2$$

with respect to $\mathbf{b} = (b_0, b_1, \dots, b_p)$.

The *least squares estimator (LSE)* $\hat{\boldsymbol{\beta}}$ is given by the solution of a system of equations known as *normal equations* 정규식.

추정대상 모수 $\boldsymbol{\beta}$ 와 잔차제곱합을 정의하는 $\mathbf{b}$ 를 구별하여 사용하고 있다.

Quantities derived by LSE

The fitted least squares regression equation is written as

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots + \hat{\beta}_p X_p.$$

The fitted values are

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_p x_{ip} \quad \text{and} \quad i = 1, 2, \dots, n.$$

The ordinary least squares residuals are given by

$$e_i = y_i - \hat{y}_i \quad \text{and} \quad i = 1, 2, \dots, n.$$

An unbiased estimator of σ^2 is given by

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n - p - 1},$$

where

$$\text{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n e_i^2.$$

Interpretations of Regression Coefficients

For SP data, the fitted LS regression equation with X_1 and X_2 is obtained as

$$\hat{Y} = 15.33 + 0.78X_1 - 0.05X_2.$$

Interpretation:

- ▶ The coefficient of X_1 suggests that each unit of X_1 adds 0.78 to Y when the value of X_2 is held fixed.
- ▶ The coefficient of X_2 suggests that each unit of X_2 subtracts about 0.05 from Y when the value of X_1 is held fixed.

This interpretation can be easily understood when we consider the fact that the multiple regression equation can be obtained from a series of simple regression equations.

A Way Obtaining $\hat{\beta}_2 = -0.05$

Procedure:

1. Regress Y on X_1 to obtain $\hat{Y} = 14.3 + 0.75X_1$ and the residuals $e_{Y \cdot X_1}$.
2. Regress X_2 on X_1 to obtain $\hat{X}_2 = 18.97 + 0.51X_1$ and the residuals $e_{X_2 \cdot X_1}$.
3. Regress $e_{Y \cdot X_1}$ on $e_{X_2 \cdot X_1}$ to obtain $\hat{e}_{Y \cdot X_1} = 0 - 0.05e_{X_2 \cdot X_1}$.
4. The coefficient of $e_{X_2 \cdot X_1}$ in this last regression is the same as the multiple regression coefficient of X_2 .

$\hat{\beta}_2 = -0.05$ represents the pure contribution of X_2 to the response Y after both variables have been linearly *adjusted for* the other predictor variables, where adjusted variables refer to $e_{Y \cdot X_1}$ and $e_{X_2 \cdot X_1}$.

Partial Regression Coefficients

편회귀계수

The same procedures can be applied to obtain the multiple regression coefficient of X_1 . Simply interchange X_2 by X_1 in the above three steps.

The regression coefficient β_j is the *partial regression coefficient*.

The simple and the multiple regression coefficients are not the same unless the predictor variables are uncorrelated.

j 번째 최제추정량 $\hat{\beta}_j$ 가 위와 같은 방식으로 계산됨은 일반적으로 증명가능하다.

▶ 적률가정하에서 다음이 성립

$\hat{\beta}_j$ are unbiased for β_j since

$$\mathbb{E}(\hat{\beta}_j) = \beta_j \text{ for } j = 0, 1, \dots, p.$$

For $0 \leq i, j \leq p$,

$$\text{Var}(\hat{\beta}_j) = \sigma^2 c_{jj}$$

$$\text{cov}(\hat{\beta}_i, \hat{\beta}_j) = \sigma^2 c_{ij}$$

where c_{ij} is the element in the i -th row and j -th column of the inverse of the corrected sum of squares and products matrix.

적률가정은 ' $\varepsilon_1, \dots, \varepsilon_n$ 들이 서로 독립이며, 모두 평균이 0이고 분산이 σ^2 인 분포에 따른다'는 가정이며, **정규성을 요구하지는 않음**에 유의한다.

For all unbiased estimates that are linear in the observations the least squares estimators have the smallest variance.

Thus, least squares estimators are said to be **BLUE** (Best Linear Unbiased Estimators).

This is known as the Gauss-Markov Theorem.

Gauss-Markov 정리는 적률가정만 요구하며 **정규성을 요구하지는 않는다**.

비교대상을 선형이면서 불편인 추정량들의 집합으로 국한할 경우에 한해 최제추정량의 분산이 최소가 된다는 뜻이며 두 가정 중에 하나라도 완화되는 경우는 최적성을 보장하지 못한다.

Sampling Distribution 표집분포

The estimator $\hat{\beta}_j, j = 0, 1, \dots, p$, is normally distributed with mean β_j and variance $\sigma^2 c_{jj}$.

$W = \frac{\text{SSE}}{\sigma^2}$ has a χ^2 distribution with $\text{df} = n - (p + 1) = n - p - 1$, and $\hat{\beta}_j$ and W (or equivalently $\hat{\sigma}^2$) are distributed independently of each other.

The vector $\hat{\boldsymbol{\beta}}$ has a $(p + 1)$ -variate normal distribution with mean vector $\boldsymbol{\beta}$ and variance-covariance matrix with elements $\sigma^2 c_{ij}$ for $i, j = 0, 1, \dots, p$.

추정량들의 표집분포는 $\varepsilon_1, \dots, \varepsilon_n$ 들의 분포가 하나의 정규분포 $N(0, \sigma^2)$ 이라는 정규성을 **요구한다**.

Multiple Correlation Coefficient

Assessment of the strength of the linear relationship between Y and the set of predictors $X_1, X_2, \dots, X_p$

- ▶ the scatter plot of Y versus $\hat{Y}$
- ▶ the correlation coefficient $\text{corr}(Y, \hat{Y})$ between Y and $\hat{Y}$

The coefficient of determination $R^2 = [\text{corr}(Y, \hat{Y})]^2$ is also given by

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (3)$$

$R = \sqrt{R^2}$ is called the *multiple correlation coefficient* because it measures the linear relationship between one variable Y and a set of variables $X_1, X_2, \dots, X_p$.

Adjusted R-squared

Definition

$$R_a^2 = 1 - \frac{\text{SSE} / (n - p - 1)}{\text{SST} / (n - 1)} \quad (4)$$

R_a^2 is known as the *adjusted R-squared* and is also used for judging the goodness of fit.

From (3) and (4) it follows that

$$R_a^2 = 1 - \frac{n - 1}{n - p - 1} (1 - R^2).$$

R_a^2 is sometimes used to compare models having different number of predictors.

Warning: Unlike R^2 , R_a^2 cannot be interpreted as the proportion of total variation in Y accounted for by the predictors.

Typical Phenomenon of $SSE(p)$, $R^2(p)$ and $R_a^2(p)$

Let $SSE(p)$, $R^2(p)$ and $R_a^2(p)$ be the ones with p predictors among a set of predictors.

Typical behavior of them is displayed in Figure 3.A.

We choose the model having the maximum value of $R_a^2(p)$.

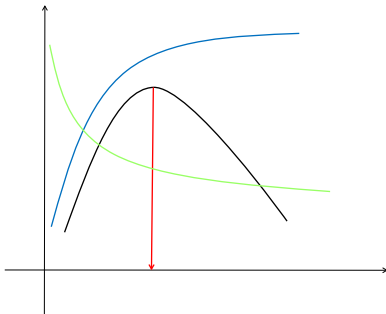


Figure 3.A: Green = $SSE(p)$, Blue = $R^2(p)$, Black = $R_a^2(p)$

Question on the adjusted R_a^2

Suppose we have two models M_p and M_q , where $q - p$ predictors are added to M_p .

Suppose $p \ll q$ and $\text{SSE}(p) \approx \text{SSE}(q)$.

Extra $q - p$ of predictors do not explain the response well since $\text{SSE}(p) \approx \text{SSE}(q)$.

Hence, there is no point to use extra large number $q - p$ of predictors.

This is so called the *principle of parsimony* 간결성원리.

Can we use adjusted R_a^2 for model selection?

간결성 원리는 반응변수를 설명하는 데에 크게 기여하지 않는 변수들은 제거하는 것이 좋다는 원리로 빅데이터 시대에도 많은 사람들이 따르는 원리이다.

Yes! Choose a model with larger R_a^2 .

Observe

$$\begin{aligned} R_a^2(p) &= 1 - \frac{n-1}{n-p-1} \frac{\text{SSE}(p)}{\text{SST}} \\ &\approx 1 - \frac{n-1}{n-p-1} \frac{\text{SSE}(q)}{\text{SST}} \\ &> 1 - \frac{n-1}{n-q-1} \frac{\text{SSE}(q)}{\text{SST}} \\ &= R_a^2(q). \end{aligned}$$

Hence, we prefer M_p to M_q .

Following the principle of parsimony, we select the model having the largest value of adjusted R_a^2 .

Testing and Confidence Limits for β_j

Testing $H_0 : \beta_j = \beta_j^0$ vs $H_1 : \beta_j \neq \beta_j^0$

t test statistic

$$T_j = \frac{\hat{\beta}_j - \beta_j^0}{\text{SE}(\hat{\beta}_j)}$$

has a t -distribution with d.f. $(n - p - 1)$, where $\text{SE}(\hat{\beta}_j) = \hat{\sigma} \sqrt{c_{jj}}$.

The confidence limits for β_j with confidence coefficient α

$$\hat{\beta}_j \pm t_{(n-p-1, \alpha/2)} \text{SE}(\hat{\beta}_j).$$

여기서도 t 통계량은 $\frac{O-E}{SE}$ 공식에 부합한다.

Regression Output for the SP Data

The fitted regression equation is

$$\hat{Y} = 10.787 + 0.613X_1 + 0.073X_2 + 0.320X_3 \\ + 0.081X_4 + 0.038X_5 - 0.038X_6.$$

Variable	Coefficient	SE	<i>t</i> -test	<i>p</i> -value
Constant	10.787	11.5890	0.93	0.3616
X_1	0.613	0.1601	3.81	0.0009
X_2	-0.073	0.1357	-0.54	0.5956
X_3	0.320	0.1685	1.90	0.0699
X_4	0.081	0.2215	0.37	0.7155
X_5	0.038	0.1470	0.26	0.7963
X_6	-0.217	0.1782	-1.22	0.2356
$n = 30$	$R^2 = 0.73$	$R_a^2 = 0.66$	$\hat{\sigma} = 7.068$	d.f. = 23

Table: Regression Output for the SP Data (Table 3.5)

Interpretation of the Output for SP Data

Constant is statistically insignificant.

- ▶ Unless there is strong theoretical reason, a constant term should always be included even if it is insignificant.
- ▶ It represents the base or background level of the response variable.

Insignificant predictors should not be in general retained.

- ▶ P-values for X_2 , X_4 and X_5 are relatively very large so that they appear to be insignificant.
- ▶ One may want to remove X_2 , X_4 , X_5 .

Types of Hypothesis in Multiple Linear Model

- A. All the regression coefficients associated with the predictors are zero.
- S. Some of the regression coefficients are zero.
- E. Some the regression coefficients are equal to each other.
- C. The regression parameters satisfy certain specified constraints.

모형간 내포성이 없는 경우에는 통계이론이 따로 정립되어 있지 않지만 상대적으로 우월한 모형을 선택하는 기법은 다양하게 존재한다. 최근 각광을 받는 방법론 중의 하나가 lasso이다.

Basic Idea of Testing in Multiple Linear Model

Definitions:

- ▶ The model given in (1) is referred to as the *full model*.
- ▶ The null hypothesis to be tested specifies values for some of the regression coefficients.
- ▶ When these values are substituted in the full model, the resulting model is called *reduced model*.

Each type of hypothesis can be formulated as a problem of testing

$$H_0 : \text{RM is adequate} \quad \text{vs} \quad H_1 : \text{FM is adequate}$$

when the reduced model is nested in the full model.

A set of models are said to be *nested* if they can be obtained from a larger model as special cases.

Testing Procedure

Let $\hat{y}_i$ and $\hat{y}_i^*$ be the values predicted for y_i by the full model and the reduced model, respectively.

Definitions:

- ▶ $SSE(FM) = \sum (y_i - \hat{y}_i)^2$
- ▶ $SSE(RM) = \sum (y_i - \hat{y}_i^*)^2$
- ▶ F-test statistic is defined by

$$F = \frac{[SSE(RM) - SSE(FM)] / (p + 1 - k)}{SSE(FM) / (n - p - 1)}$$

where FM has $p + 1$ parameters and there are k distinct parameters for RM .

H_0 is rejected if

$$F \geq F_{(p+1-k, n-p-1; \alpha)}.$$

분모의 자유도는 $(n - k) - (n - p - 1) = p + 1 - k$ 으로 계산된 것이다.

Testing the Hypothesis of Type A

Testing $H_0 : \beta_1 = \cdots = \beta_p = 0$ vs $H_1 : \beta_j \neq 0$ for some $1 \leq j \leq p$.

In other words, H_1 means *there exists a nonzero coefficient*.

► $RM : Y = \beta_0 + \varepsilon$

► $FM : Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon$

F -test statistic:

$$\begin{aligned} F &= \frac{[\text{SSE}(RM) - \text{SSE}(FM)] / (p + 1 - k)}{\text{SSE}(FM) / (n - p - 1)} \\ &= \frac{[\text{SST} - \text{SSE}] / p}{\text{SSE} / (n - p - 1)} \\ &= \frac{\text{SSR} / p}{\text{SSE} / (n - p - 1)} = \frac{\text{MSR}}{\text{MSE}} \end{aligned}$$

ANOVA Table the Hypothesis of Type A

ANOVA = ANalysis Of VAriance

MSR = Mean Square due to Regression

MSE = Mean Square due to Error

Source	Sum of Squares	d.f.	Mean Squares	F-test
Regression	SSR	p	$MSR = \frac{SSR}{p}$	$F = \frac{MSR}{MSE}$
Residuals	SSE	$n - p - 1$	$MSE = \frac{SSE}{n-p-1}$	

Table: ANOVA Table in Multiple Regression (Table 3.6)

$H_0 : \beta_1 = \cdots = \beta_p = 0$ vs $H_1 : \text{'there exists a nonzero coefficient'}$ is rejected if

$$F = \frac{MSR}{MSE} \geq F_{(p, n-p-1; \alpha)}.$$

The Hypothesis of Type A for SP Data

Source	Sum of Squares	df	Mean Squares	F-test
Regression	3148	6	525	10.5
Residuals	1149	23	50	

Table: ANOVA Table for SP Data (Table 3.7)

Since $F = \frac{525}{50} \approx 10.5$ and

```
> 1 - pf(10.5, 6, 23)
[1] 1.242621e-05
```

P-value is less than 0.0000125. Hence, we may reject H_0 and conclude that *not all β_j 's can be taken as zero*. In other words, the multiple linear regression model seems to be useful for SP data.

Testing Hypotheses of Type S

Principle of parsimony

- ▶ It enables us to isolate the most important variables.
- ▶ It provides us with a simpler description of the process studied, thereby making it easier to understand the process.

The null hypothesis states that $(p - q)$ specified variables have zero regression coefficients.

- ▶ p = the number of predictors in the full model
- ▶ q = the number of predictors in the reduced model

Testing a Hypothesis of Type S for SP Data

From Table 3.5, X_1 and X_3 appear to be significant whereas the other variables are not.

A natural question: Can X_1 and X_3 explain Y as adequately as the full set of six variables?

The null hypothesis to be tested is

$$H_0 : \beta_2 = \beta_4 = \beta_5 = \beta_6 = 0.$$

Then

$$FM : Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 + \varepsilon$$

and

$$RM : Y = \beta'_0 + \beta'_1 X_1 + \beta'_3 X_3 + \varepsilon'.$$

Calculation of F-statistic

Regression result

- ▶ FM: $SSE_6 = 1149$ (Table 3.7), $R_6^2 = 0.7326$ (Table 3.5)
- ▶ RM: $SSE_2 = 1254.65$, $R_2^2 = 0.708$ (Table 3.8)

F-test statistic for testing H_0 is

$$\begin{aligned} F &= \frac{(1254.65 - 1149)/(6 - 2)}{1149/(30 - 6 - 1)} \\ &= \frac{(0.7326 - 0.708)/(6 - 2)}{(1 - 0.7326)/(30 - 6 - 1)} \\ &= 0.528 \end{aligned}$$

Calculation of P-value by R

```
> F=((1254.65-1149)/(6-2))/(1149/(30-6-1))  
> F  
[1] 0.5287097  
> 1 - pf(0.528,4,23)  
[1] 0.7162766
```

Since P-value is 0.716, we can not reject H_0 .

The deletion of X_2, X_4, X_5, X_6 does not adversely affect the explanatory power of the model.

Linear Model in a Matrix Form

The linear model (3.1) in the text can be expressed as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (5)$$

where $x_{i0} = 1$ for all i and

$$\mathbf{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \mathbf{X} = \begin{bmatrix} x_{10} & x_{11} & \cdots & x_{1p} \\ x_{20} & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ x_{n0} & x_{n1} & \cdots & x_{np} \end{bmatrix}, \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}, \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

The *equivalence of (3.1) and (5)* follows from

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i = X_i \boldsymbol{\beta} + \varepsilon_i = (\mathbf{X}\boldsymbol{\beta})_i + \varepsilon_i = (\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon})_i$$

where X_i denotes the i th row of $\mathbf{X}$ and $(\mathbf{v})_i$, the i th element of a vector $\mathbf{v}$.

Assumptions

Let $k = p + 1$. Standard assumption on linear models are as follows.

- ▶ $X_{n \times k}$ is a matrix with fixed constants.
- ▶ $\beta = (\beta_0, \beta_1, \dots, \beta_p)_{k \times 1}$ is a (column) vector of unknown fixed constants β_j .
- ▶ ε satisfies

$$\mathbb{E}(\varepsilon) = \mathbf{0} \quad \text{and} \quad \text{Var}(\varepsilon) = \sigma^2 I_n, \quad (6)$$

where I_n is the $n \times n$ identity matrix.

The condition $\mathbb{E}(\varepsilon) = \mathbf{0}$ says that ε_i 's have zero mean. The condition $\text{Var}(\varepsilon) = \sigma^2 I_n$ implies that ε_i 's have constant variance and are uncorrelated. If they are normally distributed, then they are independent.

Least Squares Estimator (LSE)

The LSE $\hat{\boldsymbol{\beta}}$ is defined by

$$S(\hat{\boldsymbol{\beta}}) \leq S(\mathbf{b}) \quad \text{and} \quad \mathbf{b} \in \mathbb{R}^k \quad (7)$$

where

$$S(\mathbf{b}) = (\mathbf{Y} - \mathbf{X}\mathbf{b})'(\mathbf{Y} - \mathbf{X}\mathbf{b}), \quad (8)$$

or

$$\hat{\boldsymbol{\beta}} = \underset{\mathbf{b}}{\operatorname{argmin}} S(\mathbf{b}). \quad (9)$$

Call $r_i \triangleq y_i - b_0 - b_1 x_{i1} - \cdots - b_p x_{ip}$ the *ith residual*. Note that the *ith* component of $\mathbf{r} \triangleq \mathbf{Y} - \mathbf{X}\mathbf{b}$ is r_i so that the sum of squares of residuals is given by

$$S(\mathbf{b}) = \mathbf{r}'\mathbf{r} = \sum_{i=1}^n (y_i - b_0 - b_1 x_{i1} - \cdots - b_p x_{ip})^2.$$

Since the *E*stimator $\hat{\boldsymbol{\beta}}$ minimizes (*L*east) the sum of *S*quares, we refer it to be LSE.

Normal Equation for LSE

An LSE $\hat{\beta}$ satisfies the normal equation

$$X'X\hat{\beta} = X'Y. \quad (10)$$

This can be derived by

- ▶ Computing gradient or
- ▶ Geometric argument: projection.

Proposition 1

If $\hat{\beta}$ satisfies the normal equation (10), then $S(\hat{\beta}) \leq S(\mathbf{b})$ for any $\mathbf{b} \in \mathbb{R}^k$.

Full Rank X

If all columns of X are linearly independent, then we say that X is of full rank.

Proposition 2

If X is of full rank, then $X'X$ is non-singular and $\hat{\beta} = (X'X)^{-1}X'Y$ uniquely minimizes $S(\mathbf{b})$.

Proposition 3

Suppose X is of full rank. Then the LSE $\hat{\beta} = (X'X)^{-1}X'Y$ satisfies

$$\mathbb{E}(\hat{\beta}) = \beta \quad \text{and} \quad \text{Var}(\hat{\beta}) = \sigma^2 C$$

where

$$C = (X'X)^{-1} = \begin{bmatrix} c_{00} & c_{01} & \cdots & c_{0p} \\ c_{10} & c_{11} & \cdots & c_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ c_{p0} & c_{p1} & \cdots & c_{pp} \end{bmatrix}.$$

Fitted Values and Residuals Vectors

- ▶ Projection or Hat matrix P :

$$P = X(X'X)^{-1}X' \quad (11)$$

- ▶ Vector of fitted values $\hat{Y}$:

$$\hat{Y} = X\hat{\beta} = X(X'X)^{-1}X'Y = PY \quad (12)$$

- ▶ Vector of residuals e :

$$e = Y - \hat{Y} = Y - PY = (I_n - P)Y = QY \quad (13)$$

where $Q = I_n - P$.

- ▶ Residual sum of squares $S(\hat{\beta})$:

$$S(\hat{\beta}) = (Y - X\hat{\beta})'(Y - X\hat{\beta}) = e'e = Y'QY \quad (14)$$

Useful Properties

Proposition 4

The sum of ordinary LS residuals is zero.

Proposition 5

$$SST = SSR + SSE$$

Proposition 6

Let $\bar{y}$ be the mean of the response variable Y and $\bar{\hat{y}}$ be the mean of the fitted values. Then,

$$\text{corr}(Y, \hat{Y})^2 \equiv \frac{\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2 \sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2}} = R^2.$$

Sampling Distribution

From now on, we suppose that X is of full rank and $\varepsilon_1, \dots, \varepsilon_n$ are independent and normally distributed with mean 0 and variance σ^2 .

Proposition 7

The LSE $\hat{\beta}$ has a k -variate normal distribution with mean vector β and variance-covariance matrix $\sigma^2 C$.

Proposition 8

The quantity $W = e'e / \sigma^2 = \text{SSE} / \sigma^2$ has a χ^2 distribution with $n - k$ degrees of freedom.

Proposition 9

$\hat{\beta}$ and W are independent.

t statistic

Let

$$\hat{\sigma} = \sqrt{\frac{\mathbf{e}'\mathbf{e}}{n-k}} = \sqrt{\frac{\text{SSE}}{n-k}}$$

and, for $j = 0, 1, \dots, p$,

$$\text{SE}(\hat{\beta}_j) = \hat{\sigma} \sqrt{c_{jj}}.$$

Proposition 10

The test statistic

$$T_j = \frac{\hat{\beta}_j - \beta_j}{\text{SE}(\hat{\beta}_j)}$$

has a t distribution with $n - k$ degrees of freedom, where $j = 0, 1, \dots, p$.

Distribution of t -statistic

Definition 1

A random variable T has a t -distribution with $\text{df} = m$ or $T \sim t_m$ if $T = Z / \sqrt{V/m}$ where $Z \sim N(0, 1)$, $V \sim \chi_m$, and Z and V are independent.

Since $\hat{\beta}_j$ and $W = \text{SSE} / (n - k)$ are independent and

$$\frac{N(0, 1)}{\sqrt{\frac{\chi_{n-k}^2}{n-k}}} = \frac{\frac{\hat{\beta}_j - \beta_j}{\sigma c_{jj}}}{\sqrt{\frac{\text{SSE} / \sigma^2}{n-k}}} = \frac{\hat{\beta}_j - \beta_j}{\sqrt{\frac{\text{SSE}}{n-k} c_{jj}}} = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma} c_{jj}} = T_j \sim t_{n-k}.$$

Predicted value

The future y_0 corresponding to an observation vector $\mathbf{x}_0$ is

$$y_0 = \mathbf{x}_0' \boldsymbol{\beta} + \varepsilon_0, \quad (15)$$

where ε_0 is independent of $\varepsilon_1, \dots, \varepsilon_n$. Hence, y_0 can be estimated by

$$\hat{y}_0 = \mathbf{x}_0' \hat{\boldsymbol{\beta}}. \quad (16)$$

Note that

$$\mathbb{E}(y_0 - \hat{y}_0) = \mathbb{E}(y_0) - \mathbb{E}(\hat{y}_0) = \mathbf{x}_0' \boldsymbol{\beta} - \mathbf{x}_0' \boldsymbol{\beta} = 0.$$

The variance of $y_0 - \hat{y}_0$ is given by

$$\text{Var}(y_0 - \hat{y}_0) = \text{Var}(y_0) + \text{Var}(\hat{y}_0) = \sigma^2 + \sigma^2 \mathbf{x}_0' \mathbf{C} \mathbf{x}_0$$

and the standard error $\text{SE}(\hat{y}_0)$ is an estimate of $\sqrt{\text{Var}(y_0 - \hat{y}_0)}$ which is given by

$$\text{SE}(\hat{y}_0) = \hat{\sigma} \sqrt{1 + \mathbf{x}_0' \mathbf{C} \mathbf{x}_0}.$$

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